

Knight Moves

Implicit Graphs, Board BFS · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★★ Advanced

Problem

On an $N \times N$ board a knight moves in its eight L-shapes. Find the minimum number of moves from a start square to a target square.

The board squares form an implicit graph; BFS over the knight's moves gives the shortest move count.

What to Compute

- Moves: the minimum number of knight moves from (sr,sc) to (tr,tc) , or -1 if unreachable.

Input / Output Format

Input: Line 1: N . Line 2: $sr\ sc\ tr\ tc$ (0-indexed squares).

Output: One line: Moves.

Examples

Example 1

Input: 8 0 0 7 7

Output:

Moves: 6

Explanation: A knight needs six moves to cross from one corner of a standard 8x8 board to the opposite corner.

Example 2

Input: 5 0 0 4 4

Output:

Moves: 4

Explanation: On a 5x5 board the opposite corner is four knight moves away.

Constraints

- Squares are 0-indexed; both coordinates are in $[0, N-1]$.

- The knight has its eight standard L-shaped moves.
- Each move costs one; the start is 0 moves.

Sample Run

Sample Input

```
8
0 0 7 7
```

Sample Output

```
Moves: 6
```